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COURSE: AIDL

ASSIGNEMENT-5

TASK:

Lab/Exp 5: Applying Optimizers -Adam, SGD and Observing Convergence.

Kaggle Dataset Link: https://www.kaggle.com/datasets/hojjatk/mnist-dataset

Tasks:

1. Load and preprocess the MNIST dataset/example.
2. Build the required PyTorch model/program.
3. Train/execute the model.
4. Evaluate using suitable metrics.
5. Visualize results using plots where applicable.
6. Write a brief conclusion comparing observations.

```python
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader
from torchvision import datasets, transforms
import matplotlib.pyplot as plt
import numpy as np
torch.manual_seed(42)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)

transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.1307,), (0.3081,))
])
train_dataset = datasets.MNIST(
    root="./data",
    train=True,
    transform=transform,
    download=True
)
test_dataset = datasets.MNIST(
    root="./data",
    train=False,
    transform=transform,
    download=True
)

batch_size = 128

train_loader = DataLoader(
    train_dataset,
    batch_size=batch_size,
    shuffle=True
)
test_loader = DataLoader(
    test_dataset,
    batch_size=batch_size,
    shuffle=False
)
print("Training samples:", len(train_dataset))
print("Testing samples :", len(test_dataset))

fig, axes = plt.subplots(2, 5, figsize=(10, 5))

for i, ax in enumerate(axes.flat):
    image, label = train_dataset[i]
    image = image * 0.3081 + 0.1307

    ax.imshow(image.squeeze(), cmap="gray")
    ax.set_title(f"Label: {label}")
    ax.axis("off")

plt.tight_layout()
plt.show()
class MNIST_Net(nn.Module):
    def __init__(self):
        super(MNIST_Net, self).__init__()

        self.network = nn.Sequential(
            nn.Flatten(),
            nn.Linear(28 * 28, 128),
            nn.ReLU(),
            nn.Linear(128, 64),
            nn.ReLU(),
            nn.Linear(64, 10)
        )
    def forward(self, x):
```

```
    def forward(self, x):
        return self.network(x)
def train_model(optimizer_name, epochs=5):
    model = MNIST_Net().to(device)
    criterion = nn.CrossEntropyLoss()
    if optimizer_name == "SGD":
        optimizer = optim.SGD(
            model.parameters(),
            lr=0.01
        )
    elif optimizer_name == "Adam":
        optimizer = optim.Adam(
            model.parameters(),
            lr=0.001
        )
    else:
        raise ValueError("Unknown optimizer")
    train_losses = []
    train_accuracies = []
    print("\n===================================")
    print("Training using", optimizer_name)
    print("===================================")
    for epoch in range(epochs):
        model.train()
        running_loss = 0.0
        correct = 0
        total = 0
        for images, labels in train_loader:
            images = images.to(device)
            labels = labels.to(device)
            optimizer.zero_grad()
            outputs = model(images)
            loss = criterion(outputs, labels)
            loss.backward()
            optimizer.step()
            running_loss += loss.item()
            _, predicted = torch.max(outputs.data, 1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()

        epoch_loss = running_loss / len(train_loader)
        epoch_accuracy = 100 * correct / total

        train_losses.append(epoch_loss)
        train_accuracies.append(epoch_accuracy)

        print(
            f"Epoch [{epoch+1}/{epochs}] "
            f"Loss: {epoch_loss:.4f} "
            f"Accuracy: {epoch_accuracy:.2f}%"
        )

    return model, train_losses, train_accuracies
sgd_model, sgd_losses, sgd_accuracies = train_model(
    "SGD",
    epochs=5
)
adam_model, adam_losses, adam_accuracies = train_model(
    "Adam",
    epochs=5
)
def evaluate_model(model):
```

```python
    model.eval()
    correct = 0
    total = 0
    with torch.no_grad():
        for images, labels in test_loader:
            images = images.to(device)
            labels = labels.to(device)
            outputs = model(images)
            _, predicted = torch.max(outputs.data, 1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()
    accuracy = 100 * correct / total
    return accuracy
sgd_test_accuracy = evaluate_model(sgd_model)
adam_test_accuracy = evaluate_model(adam_model)

print("\n===================================")
print("FINAL TEST RESULTS")
print("===================================")

print(f"SGD  Test Accuracy : {sgd_test_accuracy:.2f}%")
print(f"Adam Test Accuracy : {adam_test_accuracy:.2f}%")
epochs_range = range(1, 6)

plt.figure(figsize=(8, 5))

plt.plot(
    epochs_range,
    sgd_losses,
    marker="o",
    label="SGD"
)

plt.plot(
    epochs_range,
    adam_losses,
    marker="o",
    label="Adam"
)

plt.xlabel("Epoch")
plt.ylabel("Training Loss")
plt.title("Convergence Comparison: SGD vs Adam")
plt.legend()
plt.grid()
```

```python
plt.show()
plt.figure(figsize=(8, 5))
plt.plot(
    epochs_range,
    sgd_accuracies,
    marker="o",
    label="SGD"
)
plt.plot(
    epochs_range,
    adam_accuracies,
    marker="o",
    label="Adam"
)
plt.xlabel("Epoch")
plt.ylabel("Training Accuracy (%)")
plt.title("Training Accuracy: SGD vs Adam")
plt.legend()
plt.grid()
plt.show()
====================================
optimizers = ["SGD", "Adam"]
accuracies = [sgd_test_accuracy, adam_test_accuracy]
plt.figure(figsize=(7, 5))
plt.bar(optimizers, accuracies)
plt.xlabel("Optimizer")
plt.ylabel("Test Accuracy (%)")
plt.title("Test Accuracy Comparison")
for i, value in enumerate(accuracies):
    plt.text(
        i,
        value + 0.2,
        f"{value:.2f}%",
        ha="center"
    )
plt.ylim(0, 100)
plt.show()
def show_predictions(model):
    model.eval()
    images, labels = next(iter(test_loader))
    images = images.to(device)
    labels = labels.to(device)
    with torch.no_grad():
        outputs = model(images)
        _, predictions = torch.max(outputs, 1)
```

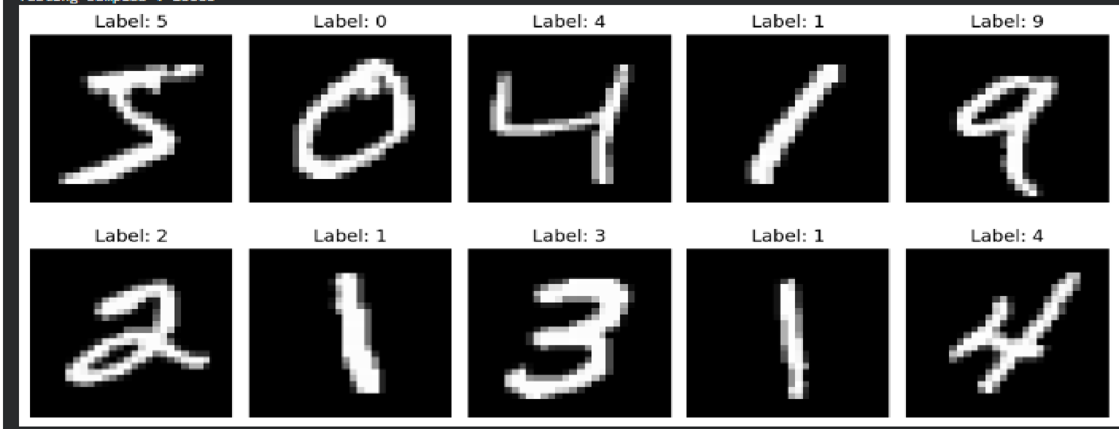
```python
    images = images.cpu()
    predictions = predictions.cpu()
    labels = labels.cpu()
    plt.figure(figsize=(12, 6))
    for i in range(10):
        plt.subplot(2, 5, i + 1)
        image = images[i] * 0.3081 + 0.1307
        plt.imshow(image.squeeze(), cmap="gray")
        plt.title(
            f"Pred: {predictions[i].item()}\n"
            f"Actual: {labels[i].item()}"
        )
        plt.axis("off")
    plt.tight_layout()
    plt.show()
print("\nSGD Predictions:")
show_predictions(sgd_model)
print("\nAdam Predictions:")
show_predictions(adam_model)
print("\n===================================")
print("CONCLUSION")
print("===================================")
if adam_test_accuracy > sgd_test_accuracy:
    print(
        "Adam achieved higher test accuracy than SGD in this experiment."
    )
else:
    print(
        "SGD achieved higher or equal test accuracy than Adam in this experiment."
    )

print(
    "Adam generally converges faster because it adapts the learning rate "
    "for individual parameters."
)

print(
    "SGD is simpler and can also achieve good performance with an appropriate "
    "learning rate."
)

print(
    "The loss and accuracy plots show the convergence behavior of both "
    "optimizers over the training epochs."
)
```

```
Using device: cpu
Training samples: 60000
Testing samples : 10000
```




```
==================================
Training using SGD
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Epoch [1/5] Loss: 1.1616 Accuracy: 70.66%
Epoch [2/5] Loss: 0.4035 Accuracy: 89.08%
Epoch [3/5] Loss: 0.3257 Accuracy: 90.70%
Epoch [4/5] Loss: 0.2880 Accuracy: 91.83%
Epoch [5/5] Loss: 0.2617 Accuracy: 92.51%

==================================
Training using Adam
==================================
Epoch [1/5] Loss: 0.3246 Accuracy: 90.64%
Epoch [2/5] Loss: 0.1338 Accuracy: 96.06%
Epoch [3/5] Loss: 0.0939 Accuracy: 97.09%
Epoch [4/5] Loss: 0.0691 Accuracy: 97.86%
Epoch [5/5] Loss: 0.0549 Accuracy: 98.27%

==================================
FINAL TEST RESULTS
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```

```
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SGD  Test Accuracy : 93.13%
Adam Test Accuracy : 97.50%
```
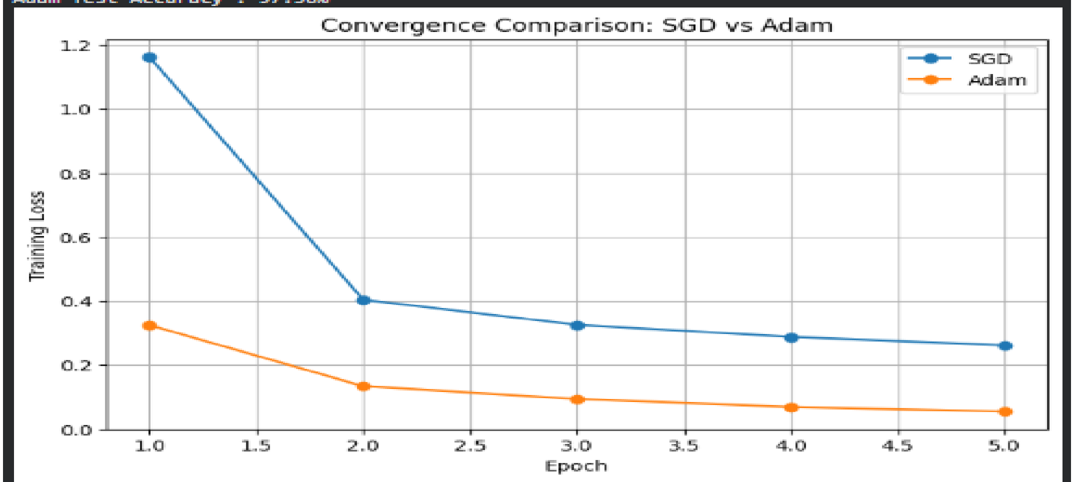
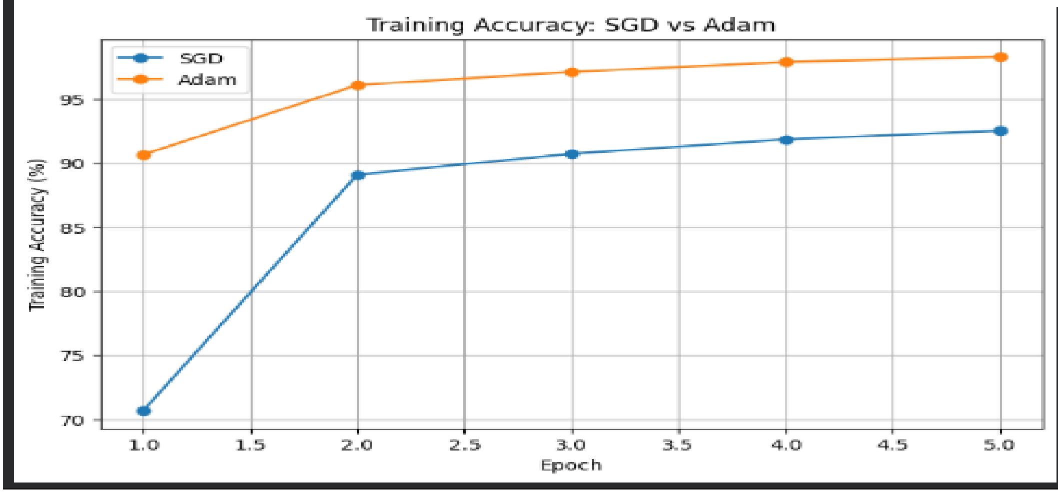
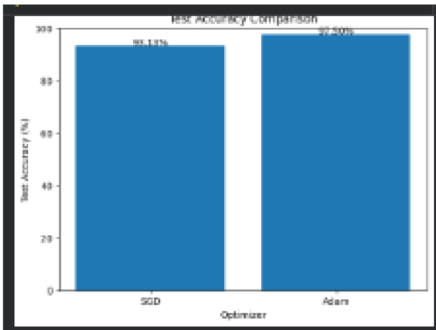
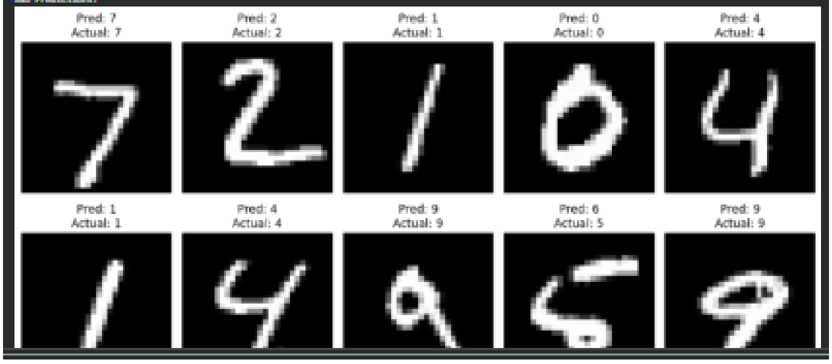



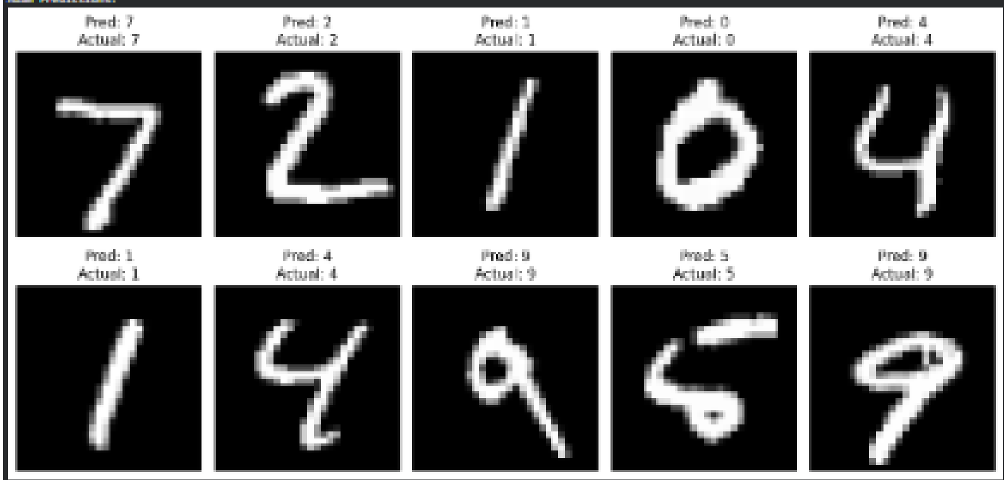




SGD Predictions:



Adam Predictions:




```
==================================================
CONCLUSION
==================================================
Adam achieved higher test accuracy than SGD in this experiment.
Adam generally converges faster because it adapts the learning rate for individual parameters.
SGD is simpler and can also achieve good performance with an appropriate learning rate.
The loss and accuracy plots show the convergence behavior of both optimizers over the training epochs.
```